Python Automata Library (PAL) Cheat Sheet

Full details: API Guide · API Reference

import PythonAutomataLibrary as pal · Safe mode is default. Call **pal.FastMode**() before constructing any PAL object. Put substantive model kernels under **@pal.njit(cache=True)**.

Create state

NewGrid: pal.NewGrid(dims,dtype): values at lattice sites

NewAgentGrid: pal.NewAgentGrid(dims,numAgentProps=0,isStackable=False): individual agents

NewPopGrid: pal.NewPopGrid(dims,capacity=None): integer counts

NewPDEgrid: pal.NewPDEgrid(dims): continuous field

NewIList: pal.NewIList(): mutable integer list

NewMultinomial: pal.NewMultinomial(): binomial/multinomial draws

Geometry / indexing

Dimensions: dims=(nx,), (nx,ny), or (nx,ny,nz); negative sizes wrap

Coordinates: grid[i] linear index; grid[x,y,z] site coordinates

Conversions: ToI(x,...); ItoX/Y/Z(i)

Neighbors: Hood(hood,x,...); pal.MooreHood, VonNeumannHood, CircleHood

Region: Box(...) iterates a rectangular coordinate region

AgentGrid

Birth: NewAgentSQ(x,...) lattice; NewAgent(x,...) continuous

Move: MoveSQ(a,x,...) lattice; Move(a,x,...) continuous

Properties: agents[a,p] reads/writes property p

Position: I(a); XSQ/YSQ/ZSQ(a); X/Y/Z(a)

Lifetime: Alive(a); Dispose(a); GetPop()

Queries: AgentsAt(...); LastAgent(...); AgentsInRadius(...)

Iteration: All(shuffle=True) snapshots living agent handles

Population / fields

State: grid[i] = v writes immediately

Queued change: Add(v,...) queues; Update() applies; Reset() clears

Population: pop.GetPop() total; pop.All() occupied indices

PDE spacing: field.SetTimeSpaceStep(dt,dx,dy,dz)

Diffusion: Diffusion, DiffusionMask, DiffusionField, DiffusionInterfaces, DiffusionADI

Radial: DiffusionRadialCircle / DiffusionRadialSphere

Advection: Advection, AdvectionField, AdvectionInterfaces

Random / lists

RNG: pal.Seed(seed); pal.Random(); pal.RandInt(n)

IList: Append, Clear, Random, Shuffle, All (copy), Iter (no copy)

Sampling: m.Binomial(n,p); m.Setup(...); m.Sample(...)

Drawing / execution

Compile: @pal.njit(cache=True) for model kernels

Mode: Safe by default; pal.FastMode() before object creation

Pixels: pal.StartPixWindow(...); pix[x,y] = color; win.Update()

OpenGL: pal.StartOpenGLWindow(...); draw.Circle/Box/Line(...)

Output: win.Save(path); StartGif, AddGifFrame, StopGif; Close